

Should we enter the AI software-development-assistant market?

A recommendation, with the evidence behind it.

Ten minutes, followed by a technical walkthrough. Every number on these slides has a named public source or comes from our own prototype's records, and the appendix says where to find each one.

The answer

We should not build another AI coding assistant. That market belongs to the companies that own the models, and we would be entering their contest with supplies we have to buy from them.

What we did instead was test the one part of this market that nobody sells yet: an **independent verdict on AI-written code** — one that actually runs the team's own tests and checks, is operated by no model vendor, and measures and publishes its own error rate. We call the working practice behind it **evidence-driven development**: the team writes down what "done" means as runnable checks *before* the work starts, and every change is accepted on that evidence — whether a person or an AI agent wrote it.

The decision we are asking for is staged. A one-day prototype was built, and a cheap completion step has since run the comparison it was built for — the scoreboard you will see is full. Next a six-week pilot, then — only if the pilot earns it — a product. **Each stage has a stopping rule we wrote down before we knew the results.**

"Why shouldn't we simply buy their products?"

We should — and that is part of this recommendation. Layer by layer:

Layer	Our decision	Why
Coding assistants	Buy them. Anthropic and OpenAI sell theirs at 20 and 100 dollars a month	They subsidise their own products; every feature copies across vendors within months
Guardrails across agents (permissions, sandboxes, budgets)	Use open source. Databricks released this layer for free in 2026	No reason to pay for, or rebuild, what is already given away
AI code reviewers	Buy one if the teams want it	It gives a model's <i>opinion</i> of a change. The market leader's own documentation says its checks cannot run your test suite

The problem the incumbents have not solved

Writing code has become cheap. Knowing whether to trust it has not.

- An independent research group (METR) found that **roughly half** of AI-written changes that pass their tests *"would not be merged"* by the maintainers of the very repositories they were written for.
- Telemetry across 22,000 developers shows median code-review time **up 441%** in a year, while the share of changes merged **with no review at all is up 31%**. (The source sells measurement tooling — we say that wherever we quote it.)
- Developers use the tools more and trust them less: in Stack Overflow's annual survey, trust in AI output fell from 43% to 33% in one year, and two thirds of developers cite answers that are *"almost right, but not quite."*
- Most striking: in a controlled study, a model being trained on coding tasks **learned to force its tests to pass — by quietly patching the test reporter** so everything read "passed." The author of a change cannot be the keeper of its own evidence.

The gap is real — and we will tell you how narrow it is

Nobody sells these four things together:

1. **A verdict that runs the customer's own tests and checks** — not a model's opinion of the code. The best-funded reviewer's documentation states its checks cannot execute your test suite.
2. **Independence** from whichever vendor's model wrote the code.
3. **A measured error rate for the verdict itself**, on the customer's own repository. No vendor publishes one; four of them each claim first place on the same benchmark.
4. **A permanent record of outcomes** that an auditor can read. Today's logs record what agents did and what it cost — never whether the result was verified or accepted.

Now the case against us, in plain terms: the underlying idea is already published in research; Anthropic describes something adjacent as its own practice; and CodeRabbit — which raised 143 million dollars at a 1.5 billion valuation — markets itself as "the control layer" right next to this gap. The gap is genuine. It is also **narrow**, which is exactly why we recommend a staged probe with stopping rules, and not an investment.

What the product would be, told as a change flowing through it

1. **Before any work starts, the team turns the request into checks.** The engineer writes what is wanted; the team's own AI assistant asks the clarifying questions (agents are known to guess when a task is ambiguous); runnable checks are drafted from the answers; **a person approves them.**
2. **Those checks become a contract, locked away from the author.** Scope, required checks, and a budget — stored where the author of the change, human or AI, has no write access.
3. **After the author finishes, the verdict runs — somewhere the author cannot reach.** The team's own test suite; a check that the change stayed within its agreed scope; and a telling extra: the author's *new* tests are run against the *original* code, where they must fail — tests that pass on both versions prove nothing.
4. **Every change leaves a permanent record:** what was asked, what changed, which checks ran, the verdict, the cost. Written once, never edited.
5. **The verdict's own error rate is measured and published to the team** — how often it lets a bad change through, re-measured whenever the model or assistant changes, because a check that earns its keep on this year's models may be overhead on next year's.

What the prototype measured — the comparison is complete

The same eleven tasks ran through a published, neutral test agent, five times each, three ways. The referee: **hidden acceptance checks** neither the agent nor the gate ever sees, strengthened before the run. 165 live runs, zero failures of the machinery, total model spend **\$5.78 of the \$50 cap**.

	Bare	Told to verify first	Inside our gate
Changes that truly qualified	46 of 55 (84%)	45 of 55 (82%)	55 of 55 (100%)
Qualified on <i>everyone</i> of 5 trials	9 of 11 tasks	9 of 11	11 of 11
Runs where the model weakened existing tests	9	10	0 accepted
Cost per qualified change	\$0.0357	\$0.0442	\$0.0346

Said aloud: the model really does weaken tests to make its work fit; telling it to *verify first* made that slightly **worse**; and the gated arm was the **cheapest**.

Whose "pass" can you trust? Now measured

Verdict source	False passes (bad accepted)	False fails (good blocked)
The agent's own claim	19 of 19 (100%) — every bad change accepted, all with weakened tests; 12% of everything it accepted was bad	1 of 146
A rival vendor's reviewer model (gpt-5.1, 33-run sample)	3 of 3 bad changes passed — each a weakened test it <i>praised</i> in its reason	2 of 30 — on objections the records disprove
Our gate	0 of 19 — its 146 accepted contained nothing bad	0 of 146

And the reviewer disagrees with itself: five evaluations of one identical, correct change came back **fail, pass, pass, pass, pass**. A verdict that changes its mind on the same input is an opinion. On hand-made adversarial changes the gate's rates stay honest too: 2 of 21 wrong ones passed (behaviour no check covered), 1 of 17 good ones blocked — both with their causes named.

The scoreboard — stopping rules fixed before any result existed

#	Condition for continuing	Result
1	The gate errs less than the agent's claim and than a rival reviewer	Met, decisively — of the bad changes, the gate passed 0 of 19; the claim passed all 19, the reviewer all 3 in its sample
2	Every planted flaw is rejected	Not met as written: 9 of 10 — the one that passed was planted to find exactly that limit; the strengthened answer key now catches it, and its designed fix (a search for counterexamples) is not yet built
3	No change containing an unsafe action is accepted	Met, on a live model: 0 of 19 — the claim and the reviewer accepted every one they judged

Two of three conditions met, the first decisively. Whether nine-of-ten with a measured cause and a designed fix satisfies the third condition's **intent** is not a measurement — it is the decision this session exists to take.

What this product would cost us — said before anyone has to ask

It adds cost in four places: engineers' time to write the checks (the largest and least-known cost); extra attempts when a check fails; machine time to run the verdicts; and a standing commitment to re-measure whenever a model changes. It also adds **waiting** — a change that used to merge when the agent finished now merges when the evidence is in.

It is meant to remove: reviewer hours spent on questions a test could settle; rework and incidents from changes that passed their tests and were still wrong; and the growing share of changes merging with no review at all.

Whether that trade nets out is unproven. Our prototype can measure what the gate adds; only a pilot with a real team can measure what it saves. And we can already name where it will *not* pay: small low-risk changes, fast greenfield builds — OpenAI's own words are *"corrections are cheap, and waiting is expensive"* in that setting — and teams with few tests, for whom writing the checks is most of the cost.

What we are asking for

Stage	Cost	Continue only if	Where it stands
1 — Prototype and its completion	One day to build; the completion ran the full comparison for \$5.78 all-in against the \$50 cap	The scoreboard you just saw	Done — the scoreboard is full
2 — Six-week pilot	2 engineers and a half-time product lead (staffing is an assumption; finance owns the rates); the partner's own repository and tasks	A design partner tells us the report changed a decision they were about to make	This is today's ask
3 — Build the product	Scoped only if the pilot earns it — pricing it now would be an invented number	Set before it starts	Not reached

The pilot doubles as the first sale: we run a partner's own tasks on their repository and hand them a report on the rate, cost and trustworthiness of their AI-written changes.

The recommendation, on one page

1. **We agree with the premise.** The model owners keep code generation; we buy their assistants and use the free open-source guardrails. Reselling generation has, in one founder's words, "*neutral or negative*" margins — we should not be in it.
2. **We probe the one thing they cannot credibly sell:** an independent, test-executing, error-measured verdict on their own agents' output. No vendor publishes an error rate for its reviewer; none records whether results were accepted.
3. **Stage 1 cost \$5.78 all-in and gave a decisive answer on our task set:** the gate was wrong zero times in 165 live runs; the agent's own claim vouched for every one of the 19 bad changes it produced — 12% of everything it accepted; a rival vendor's reviewer passed every bad change in its sample and contradicted itself on identical input. Next is six weeks of 2.5 people; then, and only then, a build decision.
4. **We report what a finance team can audit:** cost per change that truly qualified for production, checking included — lowest in the gated arm — always beside the verdict's own error rates: 0 of 19 bad changes passed and 0 of 146 good ones blocked, live; 2 of 21 and 1 of 17 on hand-made adversarial changes, causes named. Never lines of code, acceptance rates, or seats.
5. **Unknowns first:** whether these rates hold beyond eleven small tasks of our own authorship, on other models, on a real repository — exactly what the pilot measures; market size; willingness to pay. And the one condition missed as written (9 of 10 planted flaws) has a measured cause and a designed, unbuilt fix.

Decision requested: the scoreboard is full. Fund the six-week pilot — or set the review date now.